

Exact many-body relaxation gap of a fermionic reset circuit

Class-A fermionic adaptive-circuit project
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For a fixed sequence of perfect occupation resets, we prove that the charge-neutral many-body channel gap equals the covariance decay rate $-2 \log \rho(A)$, where A is the ordered product of single-particle complement projectors. The proof uses an exact triangular hierarchy of fermionic operator degrees and requires neither Gaussian input states nor commuting measured modes. We distinguish this physical relaxation sector from the unrestricted operator-space gap.

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I. RESET CHANNEL AND NEUTRAL SECTOR

Consider N complex modes with $\{\hat{\psi}_i, \hat{\psi}_j^\dagger\} = \delta_{ij}$ and $\{\hat{\psi}_i, \hat{\psi}_j\} = 0$. At step j , measure the normalized mode $\hat{\chi}_j = \sum_i v_{ji} \hat{\psi}_i$, $v_j^\dagger v_j = 1$, and reset its occupation $\hat{N}_j = \hat{\chi}_j^\dagger \hat{\chi}_j$ to a prescribed $s_j \in \{0, 1\}$. After discarding the outcome and ancillary mode, the specified Kraus channel is

$$\begin{aligned}\tilde{\mathcal{E}}_{j,0}(\hat{\rho}) &= \hat{\chi}_j \hat{\rho} \hat{\chi}_j^\dagger + (\hat{\mathbf{1}} - \hat{N}_j) \hat{\rho} (\hat{\mathbf{1}} - \hat{N}_j), \\ \tilde{\mathcal{E}}_{j,1}(\hat{\rho}) &= \hat{\chi}_j^\dagger \hat{\rho} \hat{\chi}_j + \hat{N}_j \hat{\rho} \hat{N}_j.\end{aligned}\quad (1)$$

The Kraus completeness relation follows from $\hat{N}_j^2 = \hat{N}_j$. With $\mathcal{D}[\hat{L}](\hat{\rho}) = \hat{L} \hat{\rho} \hat{L}^\dagger - \{\hat{L}^\dagger \hat{L}, \hat{\rho}\}/2$, these are exactly $\text{Id} + \mathcal{D}[\hat{\chi}_j] + \mathcal{D}[\hat{N}_j]$ and $\text{Id} + \mathcal{D}[\hat{\chi}_j^\dagger] + \mathcal{D}[\hat{N}_j]$, respectively, rather than continuous-time exponentials.

One cycle is the composition $\tilde{\mathcal{E}} = \tilde{\mathcal{E}}_{M,s_M} \circ \dots \circ \tilde{\mathcal{E}}_{1,s_1}$, with step 1 acting first; the sequence repeats every cycle. Set

$$P_j = v_j v_j^\dagger, \quad Q_j = \mathbf{1} - P_j, \quad A = Q_M \dots Q_1. \quad (2)$$

The modes may be nonorthogonal. In particular, over-complete Wannier (OW) stabilizers need not commute [1]; their order is retained in A .

For $\bar{G}_{ab} = \text{Tr}(\hat{\rho} \hat{\psi}_a^\dagger \hat{\psi}_b)$, direct application of Eq. (1) gives

$$\begin{aligned}\bar{G}' &= Q_j \bar{G} Q_j + s_j P_j, \\ \bar{G}_{n+1} &= A \bar{G}_n A^\dagger + B, \quad \delta G_{n+1} = A \delta G_n A^\dagger.\end{aligned}\quad (3)$$

Here B is independent of the input and $\delta G = \bar{G} - G_{\text{ss}}$. Thus, for eigenvalues a_i of A , the covariance multipliers are $a_i a_j^*$. Define

$$\begin{aligned}r &= \rho(A) = \max_i |a_i|, \quad 0 < r < 1, \\ \Delta_C &= -2 \log r.\end{aligned}\quad (4)$$

All rates below are per complete cycle.
The relevant operator sector is

$$\mathcal{V}_0 = \{\hat{O} : [\hat{N}_F, \hat{O}] = 0\}, \quad \hat{N}_F = \sum_i \hat{\psi}_i^\dagger \hat{\psi}_i. \quad (5)$$

Each Kraus operator has definite charge, so $\tilde{\mathcal{E}}$ is $U(1)$ covariant and preserves $\mathcal{V}_0$. Density matrices of fixed-number Slater determinants and mixtures of number sectors belong to $\mathcal{V}_0$. Fill/empty events change particle number while preserving this property. Neutrality is an operator-space condition, not a fixed-filling constraint.

II. EXACT SPECTRAL FACTORIZATION

Local deletion rule. Choose an orthonormal mode basis beginning with $\hat{\chi}_j$ and let $\hat{B}$ involve only the other modes. From Eq. (1), for an even total operator,

$$\begin{aligned}\tilde{\mathcal{E}}_{j,s}^\dagger(\hat{B}) &= \hat{B} & (\hat{B} \text{ even}), \\ \tilde{\mathcal{E}}_{j,s}^\dagger(\hat{\chi}_j^{(\dagger)} \hat{B}) &= 0 & (\hat{B} \text{ odd}), \\ \tilde{\mathcal{E}}_{j,s}^\dagger(\hat{N}_j \hat{B}) &= s \hat{B} & (\hat{B} \text{ even}).\end{aligned}\quad (6)$$

For example, filling gives $\hat{\chi}_j \hat{N}_j \hat{B} \hat{\chi}_j^\dagger + \hat{N}_j \hat{N}_j \hat{B} \hat{N}_j = (\hat{\mathbf{1}} - \hat{N}_j) \hat{B} + \hat{N}_j \hat{B} = \hat{B}$; both terms vanish for depletion. A single measured creation or annihilation operator has zero Kraus sandwiches.

Invariant filtration. A basis of $\mathcal{V}_0$ consists of normal-ordered monomials

$$\hat{O}_{I,J} = \hat{\psi}_{i_1}^\dagger \dots \hat{\psi}_{i_p}^\dagger \hat{\psi}_{j_p} \dots \hat{\psi}_{j_1}, \quad |I| = |J| = p, \quad (7)$$

including $p = 0$. Equation (6) either preserves (p, p) , annihilates the monomial, or lowers it to $(p-1, p-1)$. A change of one-particle basis preserves these degrees. Hence the span of all monomials through degree (p, p) is invariant under $\tilde{\mathcal{E}}^\dagger$.

On the quotient by lower degrees, each index is projected by Q_j or Q_j^* . Explicitly, for the moments $F_{I,J} = \text{Tr}(\hat{\rho} \hat{O}_{I,J})$, the degree-preserving block of one reset has entries

$$H_{j,p}[I, J; K, L] = \det Q_j[I, K] \det Q_j[J, L]^*. \quad (8)$$

Here $Q_j[I, K]$ is a $p \times p$ submatrix. These determinants are the matrix entries of exterior powers. The Cauchy–Binet identity gives $\wedge^p(XY) = (\wedge^p X)(\wedge^p Y)$; applying it

in the actual reset order yields the homogeneous moment propagator over one cycle,

$$T_p = (\wedge^p A) \otimes (\wedge^p A^*), \quad p = 0, \dots, N. \quad (9)$$

The dual operator-coefficient block is its transpose, with the same spectrum. Reset-source terms couple only to lower degrees and therefore do not change the eigenvalues of this block-triangular hierarchy.

The eigenvalues of an exterior power are products of distinct one-particle eigenvalues. Consequently, counting algebraic multiplicity,

$$\text{spec}(\tilde{\mathcal{E}}|_{\mathcal{V}_0}) = \bigcup_{p=0}^N \left\{ \prod_{i \in I} a_i \prod_{j \in J} a_j^* : |I| = |J| = p \right\}. \quad (10)$$

The sets I, J are independent and may overlap; each index occurs at most once within either set. This accounts for $\sum_p \binom{N}{p}^2 = \binom{2N}{N} = \dim \mathcal{V}_0$ eigenvalues. Schur triangularization proves the exterior-power statement even when A has Jordan blocks. No Gaussian-state assumption is used; Gaussian maps provide a related general framework [2].

Gap. The $p = 0$ eigenvalue is 1. Every $p \geq 1$ product satisfies $|\Lambda| \leq r^{2p} \leq r^2$. Equality is attained at $p = 1$ by $I = J = \{i_*\}$ with $|a_{i_*}| = r$. The stationary eigenvalue is therefore simple in this sector and

$$\Delta_{q=0} \equiv -\log \max_{\Lambda \neq 1} |\Lambda| = -2 \log r = \Delta_C. \quad (11)$$

This proves equality, strengthening the upper bound obtained from quadratic spectral inclusion alone.

III. PARITY SECTORS AND SCOPE

Allow p creations and ℓ annihilations in Eq. (7), with net operator charge $q = p - \ell$. For $p + \ell$ even, the same proof

gives blocks $(\wedge^p A) \otimes (\wedge^\ell A^*)$. Every nonconstant product contains at least two factors, and the neutral block attains r^2 . Thus $\Delta_{\text{even}} = \Delta_{q=0}$; even means commuting with fermion parity, not occupying only one Hilbert-space parity block.

For completeness, the unrestricted spectrum of the specified Kraus channel requires an additional step. Let $\hat{\Pi} = (-1)^{\hat{N}_F}$ and $\mathcal{S}(\hat{O}) = \hat{\Pi} \hat{O}$, so $\mathcal{S}^2 = \text{Id}$. For even and odd Kraus operators $\hat{K}_e, \hat{K}_o$,

$$(\mathcal{S} \tilde{\mathcal{E}}_j^\dagger \mathcal{S})(\hat{O}) = \hat{K}_e^\dagger \hat{O} \hat{K}_e - \hat{K}_o^\dagger \hat{O} \hat{K}_o. \quad (12)$$

On odd monomials, this sign restores Eq. (6) with the parities of $\hat{B}$ reversed: moving an odd $\hat{B}$ through the odd Kraus operator cancels the minus sign. Similarity preserves spectra and composition, giving the same blocks with $p + \ell$ odd. Their maximal modulus is r , attained at total degree one. Hence

$$\Delta_{\text{full}} = -\log r = \frac{1}{2} \Delta_C, \quad \Delta_{\text{even}} = \Delta_{q=0} = \Delta_C. \quad (13)$$

Physical density matrices have even parity, and the preparations under discussion are also charge neutral. Their channel relaxation is governed by Eq. (11). The proof concerns a fixed finite system, exact resets, and a repeated deterministic schedule. It establishes an asymptotic spectral rate, allowing Jordan prefactors; it does not by itself bound a size-uniform mixing time or a conditional-trajectory purification time. Standalone number dephasing, imperfect feedback, and averaging different schedules require a new higher-order analysis.

[1] A. Bhuiyan, H. Pan, and C.-M. Jian, *Phys. Rev. Research* **8**, 023147 (2026).

[2] S. Bravyi, Lagrangian representation for fermionic linear optics, *Quantum Inf. Comput.* **5**, 216 (2005).